Chapter 1

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1. Let x be a nilpotent element of a ring A. Show that $1+x$ is a unit of A. Deduces that a sum of a nilpotent element and a unit is a unit.

Proof. □

2. Let A be a ring and let $A[x]$ be the univariate polynomial ring of A. Let $f = a_0 + a_1x + \cdots + a_nx^n \in A[x]$. Prove that
   (a) f is a unit in $A[x] \iff a_0$ is a unit in A and $a_1, \ldots, a_n$ are nilpotent.
   (b) f is nilpotent $\iff a_0, a_1, \ldots, a_n$ are nilpotent.
   (c) f is a zero divisor $\iff$ there exists $a \neq 0$ in A such that $af = 0$.
   (d) f is said to be *primitive* if $(a_0, a_1, \ldots, a_n) = (1)$. Prove that if $f, g \in A[X]$, then fg is primitive $\iff f$ and g are primitive.

Proof. □

3. Generalize the results of Exercise 2 to a multivariate polynomial ring $A[x_1, \ldots, x_r]$.

Proof. □

4. In the ring $A[x]$, the Jacobson radical is equal to the nilradical.

Proof. □

5. Let A be a ring and $A[[x]]$ the ring of formral power series $f = \sum_{n=0}^{\infty} a_nx^n$ with coefficients in A. Show that
   (a) f is a unit in $A[[x]] \iff a_0$ is a unit in A.
   (b) If f is nilpotent, then a_n is nilpotent for all $n \geqslant 0$. Is the converse true?
   (c) f belongs to the Jacobson radical of $A[[x]] \iff a_0$ belongs to the Jacobson radical of A.
   (d) The contraction of a maximal ideal m of $A[[x]]$ is a maximal ideal of A, and m is generated by m^c and x.
   (e) Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Proof. □

15. Let A be a ring and X be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of prime ideals of A which contain E. Prove that
   (a) if $\mathfrak{a}$ is the ideal generated by E, then $V(E) = V(\mathfrak{a}) = V(\sqrt{\mathfrak{a}})$.
   (b) $V(0) = X$, $V(1) = \emptyset$.
   (c) If $(E_i)_{i \in I}$ is any family of subsets of A, then

$$V\left(\bigcup_{i \in I} E_i\right) = \bigcap_{i \in I} V(E_i).$$

   (d) $V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cap V(\mathfrak{b})$ for an ideals $\mathfrak{a}, \mathfrak{b} \trianglelefteq A$.

Proof. □

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Proof. ∎

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Proof. ∎

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Proof. ∎

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Proof. ∎

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Proof. ∎

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Proof. ∎